

FACULTY ACTIVITIES RECORD

NAME: Zbigniew Michalewicz

PERIOD COVERED BY THIS REPORT: 1 April 1999 – 31 March 2000.

I. TEACHING (Approximate hours per week devoted this category: 15)

A. Courses

	Student Credit
No. of Preparations: Fall: 1G 1U	Contact Hrs: 3G 3U Hours Generated: 300
Spring:	(I was on leave of absence)

Teaching Evaluations: excellent!

Other Pertinent Information: I co-authored a few papers with graduate students. Currently I supervise 9 PhD students:

- Raul Gallard
- Susana Esquivel
- Guillermo Leguizamon
- Ariel Leiva
- Claudio Ochoa
- Robert Hinterding
- Steven Smith
- Krzysztof Trojanowski
- Anne Olsen

Two of them will defend their thesis in first week of May 2000.

B. Advising (number):

Freshmen: 8 Upper Classmen: 6 International: 5 Graduate: Co-op: 1
Senior Projects Supervised: 8 Theses Directed: 3 Student Organizations:

Senior Projects Supervised: 8

Thesis directed:

Rudolf Paul Wieland, A Mutable Tagging System for a
Mutualistic, Species-driven Co-evolutionary Technique for Function
Optimization, 28 July 1999.

Student Organizations:

C. Curriculum and/or Course Development:

I keep upgrading the graduate course CSCI 6111, because of rapid changes in the field of evolutionary computation. I developed and proposed a new undergraduate course, CSCI 3177 Problem Solving.

D. Special Projects (video tapes, lab facilities, etc.):

II. SCHOLARSHIP AND PROFESSIONAL DEVELOPMENT (Approximate hours per week: 35)

A. Basic and Applied Research (type, how supported, proposals submitted or pending). Describe very briefly:

Additionally, I prepared two proposals for NSF; these are:

- “Evolutionary Algorithms for Inventory Problems”, \$378,027 and
- “Test Case Generator for Parameter Optimization Problems”, approx. \$300,000.

B. Scholarship:

1. Publications and reports:

a) Refereed Journal Articles Accepted during the reporting period.

- Trojanowski, K. and Michalewicz, Z., *Evolutionary Optimization in Non-Stationary Environments*, **Journal of Computer Science & Technology**, in press.
- Śmierzchalski, R. and Michalewicz, Z., *Modeling of Ship Trajectory in Collision Situations by an Evolutionary Algorithm*, **IEEE Transactions on Evolutionary Computation**, Vol.4, No.2, 2000.
- Michalewicz, Z., Deb, K., Schmidt, M., and Stidsen, T., *Test-case Generator for Constrained Parameter Optimization Techniques*, to appear in **IEEE Transactions on Evolutionary Computation**.
- Khouja, M., Michalewicz, Z., and Satoskar, S., *A Comparison between Genetic Algorithms and the RAND Method for Solving the Joint Replenishment Problem*, to appear in **Production Planning & Control**, 2000.
- Michalewicz, Z., *The Conceptual Framework of Evolutionary Computation*, to appear in **IEEE Spectrum**, 2000.

b) Refereed Journal Articles which appeared during the reporting period.

- Koziel, S. and Michalewicz, Z., *Evolutionary Algorithms, Homomorphous Mappings, and Constrained Parameter Optimization*, **Evolutionary Computation**, Vol.7, No.1, pp.19–44, 1999.

- Michalewicz, Z., Esquivel, S., Gallard, R., Michalewicz, M., Tao, G., and Trojanowski, K., *The Spirit of Evolutionary Algorithms*, **Journal of Computing and Information Technology**, special issue on Evolutionary Computing, Vol.7, No.1, pp.1–18, 1999.
- Eiben, A.E., Hinterding, R., and Michalewicz, Z., *Parameter Control in Evolutionary Algorithms*, **IEEE Transactions on Evolutionary Computation**, Vol.3, No.2, 1999, pp.124–141.
- Fogel, D.B. and Michalewicz, Z., *Why Evolutionary Algorithms?*, **Bulletin of the European Association for Theoretical Computer Science**, No. 68, June 1999, pp. 115–133.
- Michalewicz, Z., *Seven Methods for Transforming Corporate Data into Business Intelligence*, review, **Emergence: A Journal of Complexity Issues in Organizations and Management**, Vol.1, No.2, 1999, p.190–192.

c) Refereed Proceedings Articles Accepted during the reporting period.

- Trojanowski, K. and Michalewicz, Z., *Extensions of Evolutionary Algorithms for Non-stationary Environments*, Proceedings of the 4th National Conference on Evolutionary Computation and Global Optimization, May 25–28, 1999, Potok Złoty, Poland.
- Michalewicz, Z., Deb, K., Schmidt, M., and Stidsen, T., *Evolutionary Algorithms for Engineering Applications*, Proceedings of EUROGEN'99, May 30 – June 3, 1999, Jyväskylä, Finland, pp.73–94.
- Trojanowski, K. and Michalewicz, Z., *Evolutionary Approach to Non-stationary Optimisation Tasks*, Proceedings of the 11th International Symposium on Methodologies of Intelligent Systems, Lecture Notes in Artificial Intelligence, Vol.1609, Z.W. Raś and A. Skowron (eds), Springer-Verlag, pp. 538–546, June 8–11, 1999, Warsaw, Poland.
- Trojanowski, K. and Michalewicz, Z., *Evolutionary algorithms for non-stationary environments*, Proceedings of the 8th International Symposium on Intelligent Systems (ISS'99), June 14–18, 1999, Ustronie/Wisła, Poland, pp.229–240.
- Michalewicz, Z., Deb, K., Schmidt, M., and Stidsen, T., *Towards Understanding Constrained-Handling Methods in Evolutionary Algorithms*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.581–588.
- Trojanowski, K. and Michalewicz, Z., *Searching for Optima in Non-stationary Environments*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1843–1850.
- Stidsen, T., Caprani, O., and Michalewicz, Z., *A Parabolic Operator for Parameter Optimization Problems*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1494–1500.

- Leguizamón, G. and Michalewicz, Z., *A New Version of the Ant System for Subset Problems*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1459–1464.
- Michalewicz, Z. and Michalewicz, M., *Teaching Evolutionary Algorithms*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1702–1706.
- Krink, T., Mayoh, B.H., and Michalewicz, Z., *A PATCHWORK Model for Evolutionary Algorithms with Structured and Variable Size Populations*, Proceedings of the Genetic and Evolutionary Computation Conference (GECCO'99), Orlando, Florida, July 14–17, 1999, pp.1321–1328.
- Eiben, A.E., Jansen, B., Michalewicz, Z., and Paechter, B., *Solving CSPs with Using Self-adaptive Constraint Weights: How to prevent EAs from Cheating*, Proceedings of the Genetic and Evolutionary Computation Conference (GECCO'2000), Las Vegas, Nevada, July 8–12, 2000.
- Schmidt, M. and Michalewicz, Z., *Test-Case Generator TCG-2 for Nonlinear Parameter Optimization*, Proceedings of the Congress on Evolutionary Computation, CEC'2000, San Diego, July 16–19, 2000.

d) Refereed Proceedings Articles which appeared during the reporting period.

- Trojanowski, K. and Michalewicz, Z., *Extensions of Evolutionary Algorithms for Non-stationary Environments*, Proceedings of the 4th National Conference on Evolutionary Computation and Global Optimization, May 25–28, 1999, Potok Złoty, Poland.
- Michalewicz, Z., Deb, K., Schmidt, M., and Stidsen, T., *Evolutionary Algorithms for Engineering Applications*, Proceedings of EUROGEN'99, May 30 – June 3, 1999, Jyväskylä, Finland, pp.73–94.
- Trojanowski, K. and Michalewicz, Z., *Evolutionary Approach to Non-stationary Optimisation Tasks*, Proceedings of the 11th International Symposium on Methodologies of Intelligent Systems, Lecture Notes in Artificial Intelligence, Vol.1609, Z.W. Raś and A. Skowron (eds), Springer-Verlag, pp. 538–546, June 8–11, 1999, Warsaw, Poland.
- Trojanowski, K. and Michalewicz, Z., *Evolutionary algorithms for non-stationary environments*, Proceedings of the 8th International Symposium on Intelligent Systems (ISS'99), June 14–18, 1999, Ustronie/Wisła, Poland, pp.229–240.
- Michalewicz, Z., Deb, K., Schmidt, M., and Stidsen, T., *Towards Understanding Constrained-Handling Methods in Evolutionary Algorithms*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.581–588.
- Trojanowski, K. and Michalewicz, Z., *Searching for Optima in Non-stationary Environments*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1843–1850.

- Stidsen, T., Caprani, O., and Michalewicz, Z., *A Parabolic Operator for Parameter Optimization Problems*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1494–1500.
- Leguizamón, G. and Michalewicz, Z., *A New Version of the Ant System for Subset Problems*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1459–1464.
- Michalewicz, Z. and Michalewicz, M., *Teaching Evolutionary Algorithms*, Proceedings of the Congress on Evolutionary Computation, CEC'99, Washington DC, July 6–9, 1999, pp.1702–1706.
- Krink, T., Mayoh, B.H., and Michalewicz, Z., *A PATCHWORK Model for Evolutionary Algorithms with Structured and Variable Size Populations*, Proceedings of the Genetic and Evolutionary Computation Conference (GECCO'99), Orlando, Florida, July 14–17, 1999, pp.1321–1328.

e) Refereed book chapters

- Michalewicz, Z. and Schoenauer, M., *Evolutionary Algorithms*, in *Encyclopedia of Operations Research & Management Science*, 2nd edition, **Kluwer Academic**, 1999.

f) Invited book chapters

- Xiao, J. and Michalewicz, Z., *An Evolutionary Computation Approach to Planning and Navigation*, chapter in *Soft-Computing and Mechatronics*, K. Hirota and T. Fukuda (Editors), **Physica-Verlag**, 1999.
- Michalewicz, Z. and Schoenauer, M., *Evolutionary Algorithms for Constrained Parameter Optimization Problems*, chapter in *Evolutionary Computations and Intelligent Systems*, **Gordon and Breach**, 1999.

g) Contributed book chapters

h) Other papers

2. Books, software and other education materials:

- Michalewicz, Z., *Genetic Algorithms + Data Structures = Evolution Programs*, Chinese translation, **Science Press**, Beijing, 2000.
- Michalewicz, Z., Schoenauer, M., Yao, X., and Zazala, A., *Proceedings of the Congress on Evolutionary Computation*, **IEEE Service Center**, Piscataway, NJ, Washington, DC, July 6–9, 1999, Vol.1, 2, and 3.
- Michalewicz, Z. and Fogel, D., *How to Solve It: Modern Heuristics*, **Springer-Verlag**, Berlin, 2000.

- Bäck, T., Fogel, D., and Michalewicz, Z. (Editors), *Evolutionary Computation: Basic Algorithms and Operators*, **Institute of Physics**, London, 2000.
- Bäck, T., Fogel, D., and Michalewicz, Z. (Editors), *Evolutionary Computation: Advanced Algorithms and Operators*, **Institute of Physics**, London, 2000.
- a software package available for public (over 800 users all over the world).

3. Formal presentations at technical meetings:

- seminar speaker for the Department of Mathematical Modelling, Technical University of Denmark, Lyngby, Denmark, April 22, 1999.
- invited speaker for the Third Workshop on Applied/Advanced Research in Combinatorial Optimization, Technical University of Denmark, Lyngby, Denmark, April 23–24, 1999.
- invited lecturer for the “Evolutionary Algorithms in Engineering and Computer Science: Recent Advances and Industrial Applications” (EUROGEN’99), Jyväskylä, Finland, May 30 – June 3, 1999.
- 11th International Symposium on Methodologies for Intelligent Systems, Warsaw, Poland, June 8–11, 1999.
- the Eight Intelligent Information Systems Workshop, (IIS’99), Ustronie, Poland, June 14–18, 1999.
- seminar speaker at Bording Data A/S, Unistar Division, Aarhus, Denmark, June 30, 1999.
- tutorial speaker for the Congress on Evolutionary Computation, Washington, DC, July 6–9, 1999.
- seminar speaker for the University of Buenos Aires, Argentina, September 8, 1999.
- invited speaker for the Operation Research Symposium (SIO’99), part of the 28th Jornadas Argentinas de Informática e Investigación Operativa, Buenos Aires, Argentina, September 6–10, 1999.
- invited speaker for the EVALife Workshop ’99, Interdisciplinary Concepts for the Analysis of Complex Systems, University of Aarhus, Denmark, September 20–21, 1999.
- invited speaker for the International Conference of the Chilean Computer Science Society, Santiago, Chile, November 8–13, 1999.
- visiting professor at the Departamento de Informática, Universidad Santa María, Chile, November 11–15, 1999.
- invited professor, University of San Luis, Argentina, November 16–25, 1999.

C. Other Professional or Scholarly Activities (short courses attended, consulting, summer employment, etc.) Describe very briefly:

Registered P.E.-- YES: NO: NOT APPLICABLE: XX

III. PROFESSIONAL-RELATED SERVICE (Approximate hours per week: 10)

A. Internal: (Department, College, University)

1. Committees (member or officer)

- I am the chairman, member of (too) MANY committees!

2. Administrative duties

- chairman's duties
- advising, participation in meetings, etc.

3. Other (e.g., special assignments, special programs, etc.)

B. Professional Societies

1. Indicate membership and offices held:

- senior member of the IEEE.
- executive vice-president of the IEEE Neural Network Council, January 1, 1999 – December 31, 1999.

2. Meetings attended (indicate type of participation such as speaker, session chairman, etc.):

- seminar speaker for the Department of Mathematical Modelling, Technical University of Denmark, Lyngby, Denmark, April 22, 1999.
- invited speaker for the Third Workshop on Applied/Advanced Research in Combinatorial Optimization, Technical University of Denmark, Lyngby, Denmark, April 23–24, 1999.
- invited lecturer for the “Evolutionary Algorithms in Engineering and Computer Science: Recent Advances and Industrial Applications” (EUROGEN'99), Jyväskylä, Finland, May 30 – June 3, 1999.
- 11th International Symposium on Methodologies for Intelligent Systems, Warsaw, Poland, June 8–11, 1999.
- the Eight Intelligent Information Systems Workshop, (IIS'99), Ustronie, Poland, June 14–18, 1999.
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- tutorial speaker for the Congress on Evolutionary Computation, Washington, DC, July 6–9, 1999.
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- invited speaker for the International Conference of the Chilean Computer Science Society, Santiago, Chile, November 8–13, 1999.
- visiting professor at the Departamento de Informática, Universidad Santa María, Chile, November 11–15, 1999.
- invited professor, University of San Luis, Argentina, November 16–25, 1999.

3. Special assignments (reviewer, committees, etc.):

- associate editor of the international journal Statistics and Computing (Chapman & Hall).
- editorial board of the international journal Evolutionary Computation (MIT Press).
- editorial board of the international journal Fundamenta Informaticae (IOS Press).
- foundation editor of the Journal for Universal Computer Science (Springer-Verlag).
- editorial board of the Journal of Heuristics (Kluwer Academic Publishers).
- associate editor of IEEE Transactions on Neural Networks.
- associate editor of IEEE Transactions on Evolutionary Computation.
- associate editor of the international journal *Journal of Advanced Computational Intelligence* (Fuji Press).
- member of the Advisory Board for the Series on Natural Computing, Springer-Verlag.
- member of the Training Committee of Evonet (The Network of Excellence in Evolutionary Computing), January 1, 1999 – present.
- member of the Program Committee of the 11th International Symposium on the Methodologies for Intelligent Systems, Warsaw, Poland, June 8–11, 1999.
- member of the Program Committee of the 5th International Mendel Conference on Soft Computing (MENDEL'99), Brno, Czech Republic, June 9–12, 1999.

- member of the Program Committee of the Eight Intelligent Information Systems Workshop (IIS'99), Ustronie, Poland, June 14–18, 1999.
- member of the Program Committee of the International Workshop on Soft Computing in Industry'99 (IWSCI'99), June 16–18, 1999.
- member of the International Scientific Committee of the 3rd IMACS International Multiconference on Circuits, Systems, Communications, and Computers (CSCC'99), Athens, Greece, July 4–8, 1999.
- Program Chair of the Congress on Evolutionary Computation (CEC'99), Washington, DC, July 6–9, 1999.
- member of the Program Committee of the Genetic and Evolutionary Computation Conference (GECCO'99), Orlando, Florida, July 14–17, 1999.
- member of the Program Committee of the 3rd Metaheuristics International Conference (MIC'99), July 18–22, 1999, Angra dos Reis, Brazil.
- member of the Program Committee of the 4th Online World Conference on Soft Computing in Industrial Applications (WSC4), September 21–30, 1999.
- member of the Program Committee of the Evolution Artificielle 99 – EA'99, November 3–5, 1999, Boulogne, France.
- member of the Program Committee of the Seventh International Workshop on Rough Sets, Fuzzy Sets, Data Mining, and Granular-Soft Computing, November 9–11, 1999, Ube, Yamaguchi, Japan.
- member of the Program Committee of the IWES'99: International Workshop on Emergent Synthesis, December 6–7, 1999, Kobe, Japan.
- member of the Program Committee of the FEA'2000: Third International Workshop on Frontiers in Evolutionary Algorithms, Atlantic City, NJ, February 27 – March 3, 2000.
- member of the program committee of the First International Workshop on Computational Intelligence in Economics and Finance (CIEF'2000), Atlantic City, NJ, February 27 – March 3, 2000.
- member of the Program Committee of the ACDM'2000, Fourth International Conference on Adaptive Computing in Design and Manufacture, April 26–28, 2000, Plymouth, UK.
- member of the Program Committee of the 2000 Genetic and Evolutionary Computation Conference (GECCO), July 8–12, 2000, Las Vegas.
- member of the Program Committee of the Congress on Evolutionary Computation, San Diego, CA, July 16–19, 2000.
- member of the Program Committee of the 2000 Foundations of Genetic Algorithms (FOGA-6) Workshop, Charlottesville, VA, July 21–24, 2000.

- member of the Program Committee of the International Symposium on Computational Intelligence, Kosice, Slovakia, September 6–9, 2000.
- member of the Program Committee of the ANTS'2000 – From Ant Colonies to Artificial Ants: Second International Workshop on Ant Algorithms, Brussels, Belgium, September 8–9, 2000.
- member of the Program Committee of the 6th International Conference on Parallel Problem Solving from Nature, Paris, France, September 16–20, 2000.
- member of the Program Committee of the 2nd International Conference on Rough Sets and Current Trends in Computing (RSCTC'2000), Banff Park Lodge, Banff, Canada October 16–19, 2000.

C. Public or Private Agencies (consultant, committees, etc.) Explain:

D. Off-Campus Education and Public Relations:

1. Short courses (EIT Review Class) or programs:

2. Other (invited speaker, high school or community college visitation, etc.):

- I was invited speaker on many conferences; see above

E. Other Special Activities (community service, Chamber of Commerce, etc.):

Any other accomplishments, awards, etc., which might be taken into consideration:

My newest book *How to Solve It: Modern Heuristics* (Springer-Verlag, 2000) became an instant bestseller. Many universities took it as the required text.